FROM $n = 139$ ON, OEIS A351743 IS AFFINE IN BLOCKS OF LENGTH 42: PROOF OF THE CONJECTURES OF SYCAMORE AND SIGRIST

MATH0

Abstract. Let $a(1) = 1$ and let $a(n+1)$ be the greatest divisor of $s(n) = a(1) + \cdots + a(n)$ that is prime to $a(n)$; this is OEIS A351743, and s is A355149. We prove that there are explicit natural numbers P_i, Q_i, R_i, U_i $(0 \le i < 42)$ with $a(139 + 42m + i) = P_i Y_m + Q_i$ and $s(139+42m+i) = R_i Y_m + U_i$ for all $m \ge 0$, where $Y_m = (3645^m - 1)/3644$. This proves the four conjectures recorded in the two entries in 2022: 1 and 2 occur infinitely often (D. J. Sycamore); $a(2(n+21)-1) = a(2n-1)$ and $a(2(n+21)) = 3645\,a(2n) + b(n)$ with b 21-periodic, for $n \ge 70$ (R. Sigrist); and $s(n+42) = 3645\,s(n) + c(n)$ with c 42-periodic, for $n \ge 138$ (A355149). To our knowledge the result is new; both entries still list these as conjectures. The proof is verified in Lean 4 against a frozen, fidelity-reviewed statement that math0 formalized from the OEIS entries (not part of the formal-conjectures repository).

1. Introduction

The sequence A351743 of the OEIS [8], contributed by David James Sycamore on June 7, 2022, is defined by

$$a(1) = 1, \qquad a(n+1) = \text{the greatest divisor of } s(n) \text{ that is prime to } a(n) \quad (n \ge 1),$$

where $s(n) = a(1) + \cdots + a(n)$. It begins $1, 1, 2, 1, 5, 2, 3, 5, 4, 3, 1, 28, 1, 57, 2, \ldots$, and the partial sums $s(n)$ form A355149 [9] (M. De Vlieger, R. Sigrist and D. J. Sycamore, June 21, 2022). Equivalently, $a(n+1)$ is $s(n)$ with every prime power $p^k \parallel s(n)$, $p \mid a(n)$, removed; as $a(n) \mid s(n)$, $a(n+1) = 1$ exactly when $s(n)$ and $a(n)$ have the same prime divisors. Sycamore conjectured that 1, and hence 2, occur infinitely often in a; R. Sigrist conjectured on June 20, 2022 that $a(2(n+21)-1) = a(2n-1)$ and $a(2(n+21)) = 3645\,a(2n) + b(n)$ with b 21-periodic, for $n \ge 70$; and A355149 records the conjecture $s(n+42) = 3645\,s(n) + b(n)$ with b 42-periodic, for $n \ge 138$ (the exact wording is quoted in Section 2). We prove all four (Theorem 1). They follow from an exact description of the sequence from $n = 139$ on (Theorem 7): on the m-th block of 42 consecutive indices, a and s are affine functions of $Y_m = (3645^m - 1)/3644$. In particular, for odd $n \ge 139$ the terms $a(n)$ are 42-periodic with values in $\{1, \ldots, 5\}$ (eight 1's and five 2's per period), while for even n they grow by a factor of about $3645 = 3^6 \cdot 5$ per block.

Method. We exhibit 42 quadruples (P_i, Q_i, R_i, U_i) (Table 1) such that the recursion $(a(n), s(n)) \mapsto (a(n+1), s(n+1))$ maps $(P_i W + Q_i,\ R_i W + U_i)$ to the pair of the next quadruple for *every* $W \in \mathbb{N}$, and the last one to the first one at $3645W + 1$. Each step follows from Lemma 2 and finitely many facts about explicit integers, checked by the Lean kernel; the first 139 terms are computed directly. As the steps hold for all W, no p-adic information on $3645^m - 1$ is needed.

Status and prior work. As of October 2026, A351743 (revision #43, June 20, 2022) and A355149 (revision #11, November 17, 2022) still list all four statements as conjectures. Apart from A355149, the only OEIS entry citing A351743 is A357735 [10], which says nothing about them. The math0 referees found no mention of A351743, A355149, 3645 or Sycamore in the full

Date: October 20, 2026.

text of S. Fried's papers proving OEIS conjectures [2, 3, 4, 5], and no proof or counterexample on the web, arXiv, MathOverflow, Mathematics Stack Exchange or the SeqFan list; the statements are not in the formal-conjectures repository [1]. We therefore classify the result as *new*; the structure we prove is the one Sigrist observed numerically, and our contribution is a (computer-assisted) proof that it persists.

2. Statement

The conjectures. The A351743 entry [8] contains the following lines, verbatim (we omit the A-number after each line tag):

```
%N a(1)=1. Thereafter, a(n+1) is the greatest divisor of s(n) which is prime to
a(n), where s(n) is the n-th partial sum.
%C Conjecture: a(n)=1 occurs infinitely many times, and therefore so does
a(n)=2.
%F Conjectures from _Rémy Sigrist_, Jun 20 2022: (Start)
%F - a(2*(n+21)-1) = a(2*n-1) for n >= 70,
%F - a(2*(n+21)) = 3645*a(2*n) + b(n) for n >= 70 (where b is 21-periodic).
%F (End)
```

The entry is by `_David James Sycamore_, Jun 07 2022`; the COMMENTS conjecture is unsigned, so we attribute it to him. The A355149 entry [9], "Partial sums of A351743", with offset 1, contains:

```
%F Conjecture: a(n + 42) = 3645*a(n) + b(n) for n >= 138 (where b is
42-periodic).
```

Here `a` is our s; we rename this b to c.

Theorem 1. *Let a and s be as in the introduction. Then:*

(i) $a(n) = 1$ *for infinitely many* n, *and* $a(n) = 2$ *for infinitely many* n;
(ii) $a(2(n+21)-1) = a(2n-1)$ *for every* $n \geq 70$;
(iii) $a(2(n+21)) = 3645\,a(2n)+b(n)$ *for all* $n \geq 70$, *for some* $b\colon \mathbb{N} \to \mathbb{Z}$ *with* $b(n+21) = b(n)$;
(iv) $s(n+42) = 3645\,s(n) + c(n)$ *for all* $n \geq 138$, *for some* $c\colon \mathbb{N} \to \mathbb{Z}$ *with* $c(n+42) = c(n)$.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A351743.lean`, reviewed for fidelity and frozen (SHA-256 `41c82d15...47d3`) before any proof attempt. Its definitions, verbatim (import, docstrings, comments, blank lines and theorems omitted):

```
namespace Math0.Statements.A351743
def strip : ℕ → ℕ → ℕ → ℕ
  | 0, w, _ => w
  | fuel + 1, w, v => if Nat.gcd w v = 1 then w else strip fuel (w / Nat.gcd w v) v
def coprimePart (w v : ℕ) : ℕ := strip w w v
def step (p : ℕ × ℕ) : ℕ × ℕ :=
  match p with
  | (x, y) =>
    match coprimePart y x with
    | z => (z, y + z)
def state (n : ℕ) : ℕ × ℕ := step^[n] (1, 1)
def a : ℕ → ℕ
  | 0 => 0
  | n + 1 => (state n).1
def s (n : ℕ) : ℕ := ∑ k ∈ Finset.Icc 1 n, a k
def sycamore : Prop :=
  {n : ℕ | a n = 1}.Infinite ∧ {n : ℕ | a n = 2}.Infinite
def sigristOdd : Prop :=
  ∀ n : ℕ, 70 ≤ n → a (2 * (n + 21) - 1) = a (2 * n - 1)
def sigristEven : Prop :=
```

```
  ∃ b : ℕ → ℤ, (∀ n, b (n + 21) = b n) ∧
    ∀ n : ℕ, 70 ≤ n → (a (2 * (n + 21)) : ℤ) = 3645 * (a (2 * n) : ℤ) + b n
def partialSums : Prop :=
  ∃ c : ℕ → ℤ, (∀ n, c (n + 42) = c n) ∧
    ∀ n : ℕ, 138 ≤ n → (s (n + 42) : ℤ) = 3645 * (s n : ℤ) + c n
def statement : Prop :=
  sycamore ∧ sigristOdd ∧ sigristEven ∧ partialSums
end Math0.Statements.A351743
```

Here `a` has the OEIS offset 1; `a 0 = 0` is a junk value that no conjecture reaches (and it is neither 1 nor 2), and `s n` is literally $\sum_{k=1}^{n} a(k)$. As in the entry's PARI program, `coprimePart w v` divides w by $\gcd(w, v)$ until this gcd is 1 (the fuel w suffices, as each division at least halves w), and `state n` is $(a(n+1), s(n+1))$. The file proves, among others (proofs omitted):

```
theorem coprimePart_isGreatest (w v : ℕ) (hw : 0 < w) :
    IsGreatest {d | d ∣ w ∧ Nat.Coprime d v} (coprimePart w v)
theorem a_one : a 1 = 1
theorem a_succ_isGreatest (n : ℕ) (hn : 1 ≤ n) :
    IsGreatest {d | d ∣ s n ∧ Nat.Coprime d (a n)} (a (n + 1))
```

The theorems `a_one` and `a_succ_isGreatest` are the OEIS definition word for word and determine $a(n)$ for all $n \geq 1$. Three theorems proved by `decide +kernel` compare with the OEIS data: `a_data` checks all 67 listed terms of A351743, `s_data` all 50 listed terms of A355149, and `a_example` the values $a(150), \ldots, a(153) = 473394, 1, 2840365, 2$ of the entry's last example. The four conjectures are encoded literally. In `sigristOdd` the truncated subtraction is harmless, since $2n-1 \geq 139$. The periodic sequences are integer-valued, since their signs are not specified, and the relations are stated in $\mathbb{Z}$. "b is 21-periodic" is read as $b(n+21) = b(n)$ for all $n \in \mathbb{N}$; this is equivalent to periodicity for $n \geq 70$, since only the values at $n \geq 70$ enter and they extend backwards periodically. Thus `sigristEven` says exactly that $a(2(n+21)) - 3645\, a(2n)$ depends only on n mod 21 for $n \geq 70$; likewise for c in `partialSums`.

3. Proof

The proof follows the Lean file `lean/Math0/Frontier/OEIS/A351743.lean`; we give the names of the corresponding Lean declarations. For $w \geq 1$ and $v \geq 0$ let $\mathrm{cp}(w, v)$ be the greatest divisor of w prime to v (`coprimePart`), and for $(x, y) \in \mathbb{N}^2$, $y \geq 1$, put $T(x, y) = (\mathrm{cp}(y, x),\ y + \mathrm{cp}(y, x))$ (`step`). By definition, $(a(n+1), s(n+1)) = T^n(1, 1)$ for all $n \geq 0$ (`state`, `state_snd`).

3.1. Two arithmetic lemmas.

Lemma 2 (`cp_eq`)**.** *Let $d, e \geq 1$ and $v, k \geq 0$. If $\gcd(d, v) = 1$ and $e \mid v^k$, then $\mathrm{cp}(de, v) = d$.*

Proof. The number d divides de and is prime to v. Conversely, let $x \mid de$ with $\gcd(x, v) = 1$. Then $\gcd(x, v^k) = 1$, hence $\gcd(x, e) = 1$ because $e \mid v^k$, and so $x \mid d$ and $x \leq d$. Thus d is the greatest element of $\{x : x \mid de,\ \gcd(x, v) = 1\}$, which is $\mathrm{cp}(de, v)$ by `coprimePart_isGreatest`. □

Lemma 3 (`cop_lin`)**.** *Let $L, M, g, K, k, W \geq 0$. If $g \mid L$, $\gcd(M, g) = 1$ and $K \mid g^k$, then $\gcd(LW + M, K) = 1$.*

Proof. Since $LW + M \equiv M \pmod{g}$, we have $\gcd(LW + M, g) = \gcd(M, g) = 1$, hence $\gcd(LW + M, g^k) = 1$, and $K \mid g^k$ gives the claim. □

3.2. **Rows and certified steps.** A *row* is a quadruple $r = (P, Q, R, U) \in \mathbb{N}^4$. For $W \in \mathbb{N}$ put $F_r(W) = (PW + Q,\ RW + U)$ (`F`). For rows $r = (P, Q, R, U)$ and $r' = (P', Q', R', U')$ we consider two kinds of certificates.

Definition 4 (`GoodSmall`, `GoodBig`)**.** *(S)* A *small step* from r to r' with parameters (e, g, k) means: $P = 0$, $R = eP'$, $U = eQ'$, $e \mid Q^k$, $g \mid P'$, $\gcd(Q', g) = 1$, $Q \mid g^k$, $e \geq 1$, $Q' \geq 1$, $R' = R + P'$ and $U' = U + Q'$.

(B) A *big step* from r to r' with parameters (λ, f, g, h, k) means: $P' = 0$, $R = \lambda P$, $U = \lambda Q$, $\lambda = Q'f$, $h \mid P$, $h \mid Q$, $f \mid h^k$, $g \mid P$, $\gcd(Q, g) = 1$, $Q' \mid g^k$, $Q' \geq 1$, $Q \geq 1$, $f \geq 1$, $R' = R$ and $U' = U + Q'$.

In a small step the current term $a(n) = Q$ is a constant, and $s(n) = e \cdot a(n+1)$ where e is built from the primes of Q. In a big step $s(n) = \lambda\, a(n)$ for a constant λ, and the next term is the constant Q'.

Lemma 5 (`step_small`, `step_big`, `step_good`)**.** *If there is a small or a big step from r to r', then $T(F_r(W)) = F_{r'}(W)$ for every $W \in \mathbb{N}$.*

Proof. Small step. Since $P = 0$, $F_r(W) = (Q,\ RW + U)$ and $RW + U = e\,(P'W + Q')$. Here $P'W + Q' \geq Q' \geq 1$, $e \geq 1$ and $e \mid Q^k$, and $\gcd(P'W + Q', Q) = 1$ by Lemma 3 (with $L = P'$, $M = Q'$, $K = Q$). Lemma 2 gives $\mathrm{cp}(RW + U, Q) = P'W + Q'$. The new partial sum is $RW + U + P'W + Q' = (R + P')W + (U + Q') = R'W + U'$.

Big step. Put $v = PW + Q \geq Q \geq 1$. Then $RW + U = \lambda v = Q' \cdot (fv)$. Since $h \mid P$ and $h \mid Q$ we have $h \mid v$, so $f \mid h^k \mid v^k$ and $fv \mid v^{k+1}$. By Lemma 3 (with $L = P$, $M = Q$, $K = Q'$), $\gcd(v, Q') = 1$. Lemma 2 with $d = Q'$ and $e = fv \geq 1$ gives $\mathrm{cp}(RW + U, v) = Q' = P'W + Q'$, since $P' = 0$. The new partial sum is $RW + U + Q' = R'W + U'$. □

3.3. **The table.** Let $r_i = (P_i, Q_i, R_i, U_i)$, $0 \leq i \leq 41$, be the rows of Table 1 (`rows`). For the shift $\sigma(P, Q, R, U) = (3645P,\ P+Q,\ 3645R,\ R+U)$ (`shift`) we have $F_{\sigma(r)}(W) = F_r(3645W + 1)$ for all W (`F_shift`). Let $r'_i = r_{i+1}$ for $i < 41$ and $r'_{41} = \sigma(r_0) = (0,\ 3,\ 4191876044100,\ 1150350171)$ (`nextRow`).

Lemma 6 (`good_all`)**.** *For every $i < 42$ there is a step from r_i to r'_i with the parameters of Table 1: a small step (S) for even i and a big step (B) for odd i, in both cases with $k = 4$.*

Proof. Each condition is an equality, divisibility, gcd value or inequality between explicit natural numbers below $5 \cdot 10^{12}$ (the powers Q^4, g^4, h^4 are at most 15^4). In Lean, $\forall i < 42$, $\mathrm{Good}(r_i, r'_i, \mathrm{cert}_i)$ is proved by `decide +kernel`: the kernel evaluates the decision procedure, using its built-in arithmetic on numerals. □

3.4. **The block theorem.**

Theorem 7 (`key`, `a_form`, `s_form`)**.** *Let $Y_0 = 0$ and $Y_{m+1} = 3645\,Y_m + 1$, so that $Y_m = (3645^m - 1)/3644$. For all $m \geq 0$ and $0 \leq i \leq 41$,*

$$a(139 + 42m + i) = P_i Y_m + Q_i, \qquad s(139 + 42m + i) = R_i Y_m + U_i.$$

Proof. Equivalently, $T^{138+42m+i}(1, 1) = F_{r_i}(Y_m)$. We prove this by induction on m, and for each m by induction on i.

Base. $T^{138}(1, 1) = (a(139), s(139)) = (3, 315591) = F_{r_0}(0)$ (`state_base`: `decide +kernel` evaluates the 138 steps in the kernel; the partial sums involved are at most $s(139) = 315591$).

From i to $i + 1$, for $i < 41$. By the induction hypothesis and Lemmas 5 and 6 with $W = Y_m$, $T^{138+42m+i+1}(1, 1) = T(F_{r_i}(Y_m)) = F_{r_{i+1}}(Y_m)$.

even i: $P_i = 0$, small step (S)						odd i: big step (B)								
i	Q_i	R_i	U_i	e	g	i	P_i	Q_i	R_i	U_i	λ	f	g	h
0	3	1150034580	315591	3	3	1	383344860	105197	1533379440	420788	4	1	2	1
2	4	1533379440	420792	8	2	3	191672430	52599	1725051870	473391	9	9	1	3
4	1	1725051870	473392	1	1	5	1725051870	473392	3450103740	946784	2	2	1	2
6	1	3450103740	946785	1	1	7	3450103740	946785	6900207480	1893570	2	1	2	15
8	2	6900207480	1893572	4	2	9	1725051870	473393	8625259350	2366965	5	1	5	1
10	5	8625259350	2366970	5	5	11	1725051870	473394	10350311220	2840364	6	6	1	6
12	1	10350311220	2840365	1	1	13	10350311220	2840365	20700622440	5680730	2	1	2	5
14	2	20700622440	5680732	4	2	15	5175155610	1420183	25875778050	7100915	5	1	5	1
16	5	25875778050	7100920	5	5	17	5175155610	1420184	31050933660	8521104	6	2	3	2
18	3	31050933660	8521107	3	3	19	10350311220	2840369	41401244880	11361476	4	1	2	1
20	4	41401244880	11361480	8	2	21	5175155610	1420185	46576400490	12781665	9	9	1	15
22	1	46576400490	12781666	1	1	23	46576400490	12781666	93152800980	25563332	2	2	1	2
24	1	93152800980	25563333	1	1	25	93152800980	25563333	186305601960	51126666	2	1	2	3
26	2	186305601960	51126668	4	2	27	46576400490	12781667	232882002450	63908335	5	1	5	1
28	5	232882002450	63908340	5	5	29	46576400490	12781668	279458402940	76690008	6	6	1	6
30	1	279458402940	76690009	1	1	31	279458402940	76690009	558916805880	153380018	2	1	2	1
32	2	558916805880	153380020	4	2	33	139729201470	38345005	698646007350	191725025	5	5	1	5
34	1	698646007350	191725026	1	1	35	698646007350	191725026	1397292014700	383450052	2	2	1	6
36	1	1397292014700	383450053	1	1	37	1397292014700	383450053	2794584029400	766900106	2	1	2	1
38	2	2794584029400	766900108	4	2	39	698646007350	191725027	3493230036750	958625135	5	1	5	1
40	5	3493230036750	958625140	5	5	41	698646007350	191725028	4191876044100	1150350168	6	2	3	2

TABLE 1. The rows $r_i = (P_i, Q_i, R_i, U_i)$, $0 \le i \le 41$, with $a(139 + 42m + i) = P_i Y_m + Q_i$ and $s(139 + 42m + i) = R_i Y_m + U_i$ (left: even i, where $P_i = 0$; right: odd i), and the parameters of the step $r_i \to r_i'$ (Definition 4; $k = 4$ throughout). These are the lists `rows` and `certs` of the Lean file.

From block m to block $m+1$. Since $138 + 42(m+1) = (138 + 42m + 41) + 1$, Lemmas 5 and 6 and `F_shift` give $T^{138+42(m+1)}(1,1) = T(F_{r_{41}}(Y_m)) = F_{\sigma(r_0)}(Y_m) = F_{r_0}(3645 Y_m + 1) = F_{r_0}(Y_{m+1})$. □

By Table 1, $P_i = 0$ for every even i (`row_even_const`, checked by `decide`), with

$$(Q_0, Q_2, \dots, Q_{40}) = (3, 4, 1, 1, 2, 5, 1, 2, 5, 3, 4, 1, 1, 2, 5, 1, 2, 1, 1, 2, 5),$$

whereas $P_i > 0$ and $s(n) = \lambda_i\, a(n)$ with $\lambda_i \in \{2, 4, 5, 6, 9\}$ for odd i. For instance $a(150) = Q_{11} = 473394$, $a(151) = Q_{12} = 1$, $a(152) = Q_{13} = 2840365$ and $a(153) = Q_{14} = 2$ are the values in the last example of the OEIS entry.

3.5. **Proof of Theorem 1.** *(i)* (`one_inf`, `two_inf`.) As $r_4 = (0, 1, \dots)$ and $r_8 = (0, 2, \dots)$, Theorem 7 gives $a(143 + 42m) = 1$ and $a(147 + 42m) = 2$ for all $m \ge 0$.

(ii) (`odd_part`.) Let $n \ge 70$. Then $2n - 1 \ge 139$, so $2n - 1 = 139 + 42m + i$ with $m \ge 0$ and $0 \le i < 42$, and i is even because $2n-1$ and 139 are odd. Then $2(n+21)-1 = 139+42(m+1)+i$, and by Theorem 7 and $P_i = 0$ both $a(2n-1)$ and $a(2(n+21)-1)$ equal Q_i.

(iii) (`even_part`.) For $j < 42$ put $\delta_a(j) = P_j + Q_j - 3645\, Q_j \in \mathbb{Z}$ (`dA`), and let $b(n) = \delta_a((2(n \bmod 21)+29) \bmod 42)$, so that $b(n+21) = b(n)$. Let $n \ge 70$ and write $2n = 139+42m+i$ with $m \ge 0$ and $0 \le i < 42$. Since $-139 \equiv 29 \pmod{42}$ and $2n \equiv 2(n \bmod 21) \pmod{42}$, we have $i = (2(n \bmod 21) + 29) \bmod 42$, so $b(n) = \delta_a(i)$. As $2(n+21) = 139 + 42(m+1) + i$ and $Y_{m+1} = 3645 Y_m + 1$,

$$a(2(n+21)) = P_i(3645 Y_m + 1) + Q_i = 3645\,(P_i Y_m + Q_i) + P_i + Q_i - 3645\, Q_i = 3645\, a(2n) + b(n).$$

(iv) (`sums_part`.) For $j < 42$ put $\delta_s(j) = R_j + U_j - 3645\, U_j \in \mathbb{Z}$ (`dS`), and define $c(n) = \delta_s((n+29) \bmod 42)$; then $c(n+42) = c(n)$. For $n \ge 139$ write $n = 139 + 42m + i$ with $0 \le i < 42$; then $i = (n+29) \bmod 42$, and exactly as in (iii), $s(n+42) = R_i(3645 Y_m + 1) + U_i = 3645\, s(n) + c(n)$. For $n = 138$: Theorem 7 with $m = 0$ gives $s(139) = U_0 = 315591$, $a(139) = Q_0 = 3$ and

$s(180) = U_{41} = 1150350168$, so $s(138) = 315588$; and $c(138) = \delta_s(41) = 31908$, and indeed $3645 \cdot 315588 + 31908 = 1150350168 = s(180)$. □

Remark 8 (sharpness; not formalized)**.** The thresholds are optimal: $a(179) = 5 \neq 101 = a(137)$, $a(180) - 3645\,a(138) \neq a(222) - 3645\,a(180)$ and $s(179) - 3645\,s(137) \neq s(221) - 3645\,s(179)$, so (ii)–(iv) fail with thresholds $69, 69, 137$ (checked in Python, also by the math0 referees).

4. Machine verification

Files and declarations. The proof is in `lean/Math0/Frontier/OEIS/A351743.lean` of the math0 repository, `https://github.com/Foundation-0/math0`, which imports the frozen statement file (Section 2) and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A351743.result` of type `Math0.Statements.A351743.statement`. The auxiliary declarations of this file named in Section 3, including Theorem 7, are `private`. Kernel computations (`good_all`, `state_base` by `decide +kernel`; small `decide`/`rfl` steps) add no axioms; there is no `native_decide`, `sorry`, `axiom` or `set_option`.

Audit. The file ends with the command

```
#math0_settles Math0.Frontier.A351743.result Math0.Statements.A351743.statement
```

which is used for statements that math0 formalizes itself. It fails unless (1) `statement` is a universe-monomorphic definition of type `Prop`; (2) `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound` (rejecting `sorryAx`, `Lean.ofReduceBool` from `native_decide`, and custom axioms); and (3) the type of `result` is syntactically the statement or its negation (up to metadata, no unfolding). It reports

```
math0 audit: 'Math0.Statements.A351743.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A351743.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A351743.result' proves 'Math0.Statements.A351743.statement'
```

Toolchain and reproduction. Lean 4 [7] `v4.33.1` and Mathlib [6] `v4.33.1`, pinned in `lean/`; formal-conjectures [1] is pinned at commit `c77fdd4` but not used here. To reproduce, run `scripts/verify.sh` from the repository root; it fetches the Mathlib cache, rebuilds all math0 results with `lake build` (re-running every audit) and checks that the pinned dependencies are unmodified. This file alone: `lake env lean Math0/Frontier/OEIS/A351743.lean` in `lean` (under two minutes on our machine).

References

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[2] S. Fried, *Proofs of some conjectures from the OEIS*, arXiv:2410.07237, 2024.

[3] S. Fried, *Proofs of several conjectures from the OEIS*, J. Integer Seq. **28** (2025), Article 25.4.3.

[4] S. Fried, *Proofs of ten conjectures from the OEIS*, J. Integer Seq. **29** (2026), Article 26.1.8.

[5] S. Fried, *Further proofs of conjectures from the OEIS*, arXiv:2607.24832, 2026.

[6] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[7] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[8] The OEIS Foundation Inc., `https://oeis.org/A351743`, entry by D. J. Sycamore (June 7, 2022), with conjectures by D. J. Sycamore and R. Sigrist; revision #43, June 20, 2022.

[9] The OEIS Foundation Inc., `https://oeis.org/A355149`, entry by M. De Vlieger, R. Sigrist and D. J. Sycamore (June 21, 2022); revision #11, Nov. 17, 2022.

[10] The OEIS Foundation Inc., https://oeis.org/A357735, entry by D. J. Sycamore (Oct. 11, 2022).

Produced by Claude (an AI model developed by Anthropic) in the math0 project